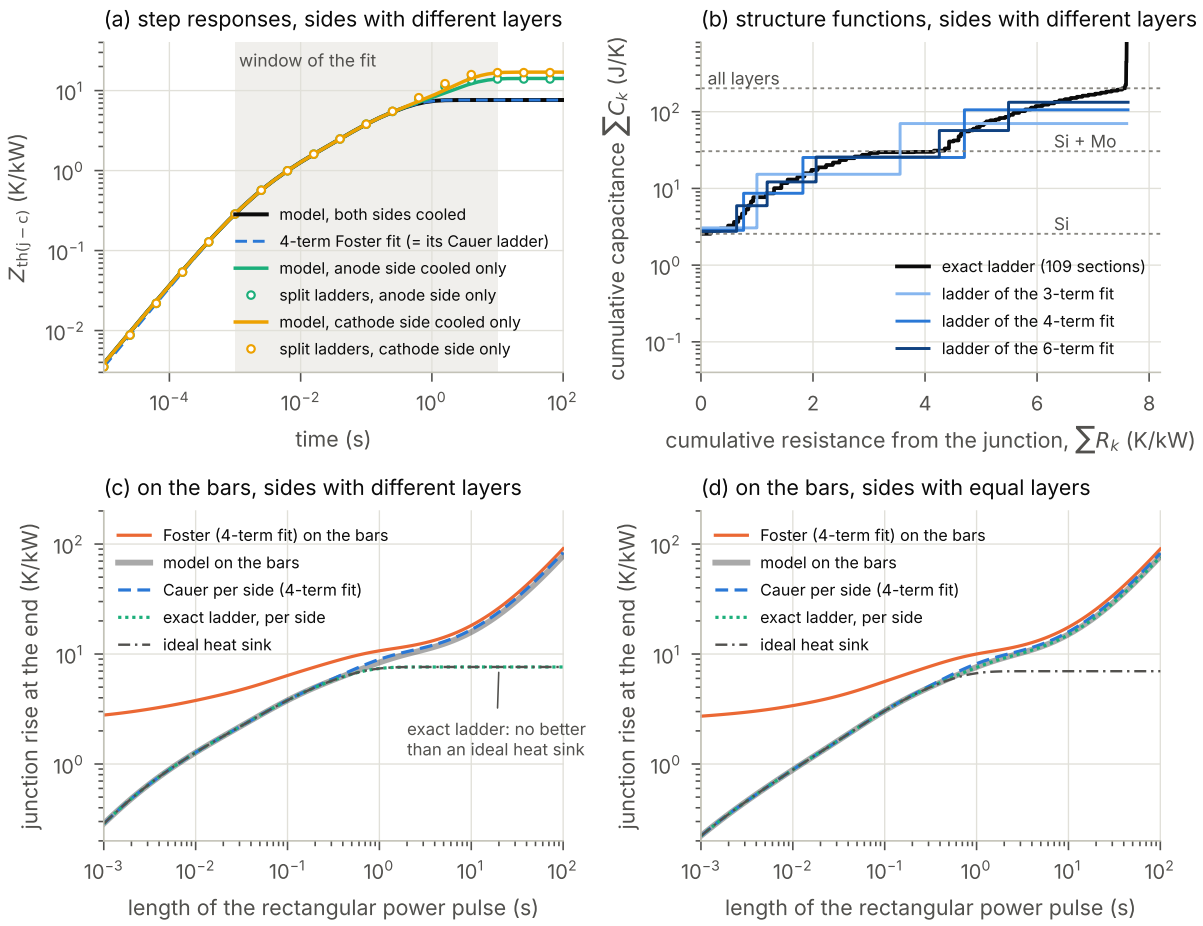
(a) step responses, sides with different layers
window of the fit
model, both sides cooled
4-term Foster fit (= its Cauer ladder)
model, anode side cooled only
split ladders, anode side only
model, cathode side cooled only
split ladders, cathode side only
$Z_{th(j-c)}$ (K/kW)
time (s)
(b) structure functions, sides with different layers
all layers
Si + Mo
Si
exact ladder (109 sections)
ladder of the 3-term fit
ladder of the 4-term fit
ladder of the 6-term fit
cumulative capacitance $\sum C_k$ (J/K)
cumulative resistance from the junction, $\sum R_k$ (K/kW)
(c) on the bars, sides with different layers
Foster (4-term fit) on the bars
model on the bars
Cauer per side (4-term fit)
exact ladder, per side
ideal heat sink
exact ladder: no better
than an ideal heat sink
junction rise at the end (K/kW)
length of the rectangular power pulse (s)
(d) on the bars, sides with equal layers
Foster (4-term fit) on the bars
model on the bars
Cauer per side (4-term fit)
exact ladder, per side
ideal heat sink
junction rise at the end (K/kW)
length of the rectangular power pulse (s)